OpenPMUT Installation and Usage Guide: Simulate Large-Scale PMUT Arrays Combining Equivalent-Circuit Models with GPU Acceleration

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1. PURPOSE AND SCOPE

The purpose of this document is to provide a step-by-step procedure for users seeking to install and run the OpenPMUT Desktop application (V1.0.0). This software simulates large-scale Piezoelectric Micromachined Ultrasonic Transducer (PMUT) arrays by combining Equivalent-Circuit Models (ECM) with multi-GPU acceleration. This SOP is explicitly written for users with an electronics background but no prior experience with Linux or GPU server management.

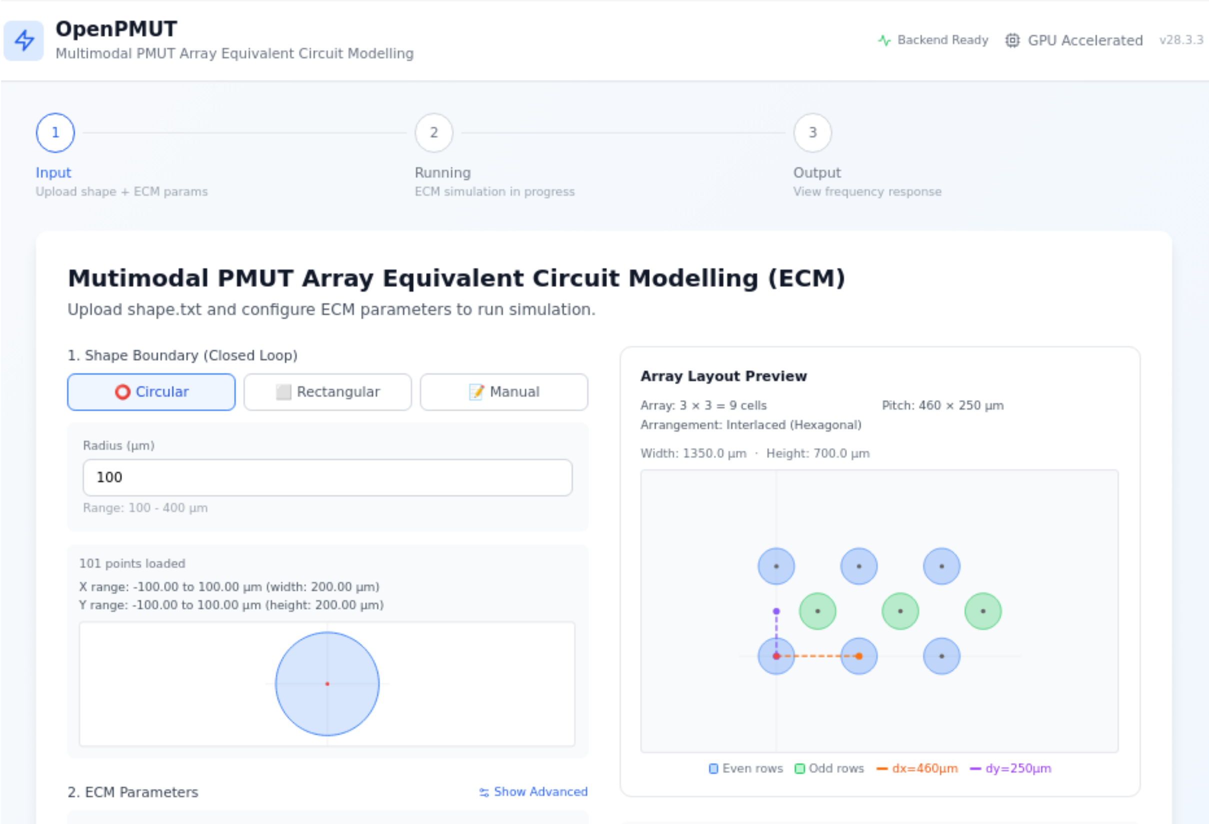

Fig. 1. PiezoMeter (PM300) test setup for measuring d33 coefficient.

Definitions and Abbreviations

PMUT: Piezoelectric Micromachined Ultrasonic Transducer. A MEMS-based ultrasonic transducer that uses piezoelectric materials to transmit and receive acoustic signals.

ECM: Equivalent-Circuit Model. A method of modeling the mechanical and acoustic behavior of the PMUT array using electrical equivalent circuits to speed up simulation.

GPU: Graphics Processing Unit. A specialized hardware component required by OpenPMUT to highly accelerate the intensive matrix calculations and eigenmode solving during simulation.

Linux Terminal / Command Line: The text-based interface used to give instructions to the Linux server.

X11 Display: A graphical windowing system for Linux. It allows the graphical user interface (GUI) of the OpenPMUT desktop app to be displayed on your personal computer screen while the heavy computations run on the remote GPU server.

Validity

The instructions regarding laboratory operation, use of IT equipment, and terms of safety must be obeyed to the best of your knowledge and belief.

2. ACCESS AUTHORISATION

Users must have an active account for the Linux GPU computing cluster and be authorized to allocate GPU resources.

3. PROCEDURE

3.1 Booking and Accessing the GPU Server

To run the simulation, you must first connect to a Linux server equipped with an NVIDIA GPU and graphical display forwarding (X11).

• Step 1: Connect to the Server with Graphical Interface support.

How to request SAL Linux account (Pitstop), see https://pitstop.research.silicon-austria.com/pitstop ; how to request ETX account, see https://itsm.silicon-austria.com

• Step 2: Open a Terminal. Once connected to the desktop environment of the server, open a "Terminal" application (often found in the Applications menu under System Tools or Utilities). You will type the commands in the following steps into this terminal window.

• Step 3: Book the GPU.

Connect to the SLURM environment, please adapt the parameters according to your requirements (for example, adding --mem-per-cpu=10G for more memory):

```
srun --partition=compute --nodes=1 --tasks-per-node=1 --cpus-per-task=2 --job-name "OpenPMUT-gpu" --time 10:00:00 --gpus-per-node=1 --pty bash
```

```
xuj@lnzlogin02 ~ :
#Book GPU and Memory
unset LD_PRELOAD
srun --partition=compute --nodes=1 --tasks-per-node=1 --cpus-per-task=32 --job-name "0311
2025" --time 10-00:00:00 --gpus-per-node=1 --pty --mem-per-cpu=4G --mail-type=TIME_LIMIT_
80 --mail-user="jiapeng.xu@silicon-austria.com" bash

*****************************************************
Loading SAL User Setup:
  - enable modules              ==> /sdf/tools/Modules/init/bash
  - load global alias file      ==> /sdf/flow/setup/alias
  - load global environment file ==> /sdf/flow/setup/env
*****************************************************
Checking workload/batch scheduler:
  ==> Slurm is active
*****************************************************
Loading private User Setup:
  - load private setup file     ==> /home/xuj/.private
  - load private alias file     ==> /home/xuj/.alias
*****************************************************
Loading default modules
*****
** SAL - Design Flow (sdf) v1.3
*****
** sdf [project] [subproject] [workspace]                ==> to enter a project
** sdf_os [os] [project] [subproject] [workspace]        ==> to enter a project with spec
ific os
** create_project [project]                              ==> to create a project area
** create_subproject <project> [subproject]              ==> to create a subproject area
** create_workspace <project> <subproject> [workspace]   ==> to create a workspace area
*****
```

3.2 Acquiring and Installing OpenPMUT

The OpenPMUT app comes pre-packaged. No complex graphical installation wizards are required.

• Step 1: Download the application package.

[Insert instructions on where the user should download the OpenPMUT-Desktop.zip file from. E.g., "Download OpenPMUT-Desktop.zip from the shared network drive / IT portal / GitHub address to your server working directory"].

• Step 2: Unzip the package. In the terminal you opened, type the following command and press Enter:

unzip OpenPMUT-Desktop.zip

• Step 3: Enter the newly created application folder by typing:

cd OpenPMUT-Desktop;

```
(saljupytertensorflow) xuj@lnzsdfml02 ~/OpenPMUT_simulation_platform : git clone https://github.com/Jasper123y/OpenPMUT.git
cd OpenPMUT
./openpmut
Cloning into 'OpenPMUT'...
remote: Enumerating objects: 200, done.
remote: Counting objects: 100% (200/200), done.
remote: Compressing objects: 100% (170/170), done.
remote: Total 200 (delta 44), reused 165 (delta 19), pack-reused 0 (from 0)
Receiving objects: 100% (200/200), 8.27 MiB | 15.80 MiB/s, done.
Resolving deltas: 100% (44/44), done.
```

3.3 Initial Setup and Python Environment Setup

The application requires a Python environment to run the backend calculations. The OpenPMUT launcher handles this automatically for you.

• Step 1: In the terminal, while inside the OpenPMUT-Desktop folder, launch the application by typing:

./openpmut

• Step 2: On the first run, the terminal script will guide you through setting up Python. Read the prompts on the screen:

If you already have a suitable environment configured, the software will detect it and start immediately.

If the software detects a Python manager (like Conda) but no environment, it will ask permission to create one for you named "openpmut". Press Enter to accept.

If you do not have Python/Conda installed at all, the script will offer to download and install Miniconda for you automatically. Follow the on-screen instructions (e.g., typing 'yes' or pressing Enter when prompted).

```
 OpenPMUT launch — Wed May 20 11:32:21 AM CEST 2026

[Setup] Checking conda availability...
[✓] conda is available: conda 4.12.0
[Setup] Scanning conda environments...
        (required: fastapi, uvicorn, torch, numpy, scipy, pandas, matplotlib, imageio, pydantic_settings)

   Conda environments:

    1) [✗] cochlea_env2                 missing: fastapi uvicorn torch imageio pydantic_settings
    2) [✗] cochlea_env3                 missing: fastapi uvicorn torch imageio pydantic_settings
    3) [✗] diffu                        missing: fastapi uvicorn pydantic_settings
    4) [✗] gnn_explainer                missing: fastapi uvicorn imageio pydantic_settings
    5) [✓] openpmut                     all packages ok ← recommended
    6) [✗] saljupyterpytorch            missing: fastapi uvicorn imageio pydantic_settings
    7) [✗] saljupyterpytorch_GNN        missing: fastapi uvicorn imageio pydantic_settings
    8) [✗] saljupyterpytorch_GNN2       missing: fastapi uvicorn pydantic_settings
    9) [✗] saljupyterpytorch_GNN3       missing: fastapi uvicorn pydantic_settings
   10) [✗] saljupyterpytorch_GNN3a      missing: fastapi uvicorn pydantic_settings
   11) [✗] saljupyterpytorch_GNN5       missing: fastapi uvicorn pydantic_settings
   12) [✗] /home/xuj/anaconda3          missing: fastapi uvicorn torch pydantic_settings
   13) [✗] /home/xuj/anaconda3/envs/new_tf_env missing: fastapi uvicorn torch pandas imageio pydantic_settings
   14) [✗] /home/xuj/anaconda3/envs/pytorchenv missing: fastapi uvicorn pandas matplotlib imageio pydantic_settings
   15) [✗] /home/xuj/anaconda3/envs/saljupyterTensorflow missing: fastapi uvicorn torch pydantic_settings
   16) [✗] /home/xuj/anaconda3/envs/saljupyterpytorch missing: fastapi uvicorn pydantic_settings
   17) [✗] (base)                       missing: fastapi uvicorn imageio pydantic_settings
   18) [✗] base                         missing: fastapi uvicorn torch pydantic_settings

    n) Create new 'openpmut' conda env with all dependencies
    q) Abort

 Environments marked [✓] have all required packages.
 Environments marked [✗] are missing packages — they will be
 installed automatically if you choose that environment.

 Choose [1-18, n, or q] (default: 5):
```

• Step 3: Once the setup completes, the script will remember your configuration for future use (saving it in a local file) so you never have to do this step again.

```
  Previously used Python environment found:
    /home/xuj/.conda/envs/openpmut/bin/python  (Python 3.11.14)

  Use this environment? [Y/n] Y
[✓] Using saved Python: /home/xuj/.conda/envs/openpmut/bin/python

  Python:  /home/xuj/.conda/envs/openpmut/bin/python (Python 3.11.14)
  Port:    18765
  Dir:     /home/xuj/OpenPMUT_simulation_platform/OpenPMUT-Desktop

[→] Verifying packages (importing torch, fastapi, etc.)... OK
[✓] All required Python packages verified.
[✓] Electron runtime found.
[✓] Application files found.

[→] Starting backend on port 18765...
[→] Waiting for backend.....
[✓] Backend ready! (6s)
[→] Launching OpenPMUT window...
```

Fig. 4. Steps for DUT preparation.

3.5 Metallization for contacting sample and piezometer

This step of preparation will be carried out at SAL's III-V fab, by trained personnel from this fab or by research staff upon basic training (not an activity critical for safety).

- Metal painting procedure: Apply silver conductive paint to the sample using a small brush to the top surface of the piezoelectric layer to create “top metallization”. Let the sample dry for about 20 minutes. Flip the sample over and paint the bottom surface of the substrate and the edge of the sample in order to create “bottom metallization and “edge metallization”, respectively. Note that the “edge metallization” provides electrical connection between the “bottom metallization” and the bottom metal electrode.

- Verification / inspection procedure: to verify the DUT is properly prepared, use a multimeter to check there is no short circuit between the top metallization and the bottom metal electrode (bottom metallization).

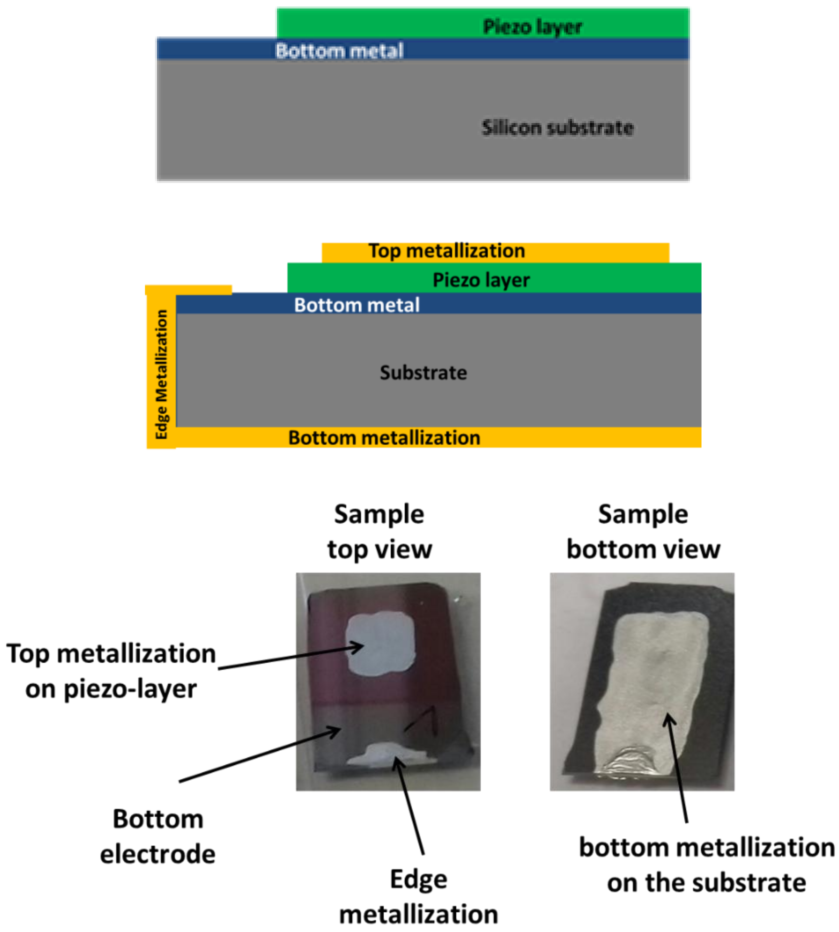

Fig. 5. DUT metallization.

3.6 d33 characterization using PM300 PiezoMeter

The table below describes the PM300 setup and measurement procedure steps:

Step	Instruction
1	Supply mains A.C. power to the PM300.
2	Switch on the PM300 and set desired parameters for: • d33 range -e.g. for AlN: 0 to 10 pC/N and Sc-AlN: 1 to 100 pC/N • Dynamic Force -e.g. for AlN: 0.25 N • Test Frequency -e.g. for AlN: 110 Hz • Static Force -e.g. for AlN: 10 N
3	Clamp the sample on the measurement head according to user manual (note that the RED light indicate too much force while GREEN light indicates the force is correct). The static force should be at the level of approximately 10N.
4	Read the d33 coefficient from the front panel of the PiezoMeter noting the measured value in the user lab book. Note that besides the d33 coefficient, the PM300 also shows the DUT capacitance and loss factor in the tool's front panel. If this does not happen and the text message "*Under*" is displayed instead, it means the DUTs may be electrically shorted between the top and bottom metallization.

Extraction of d33 from the PM300 is typically a one-time measurement for a given sample. Therefore, it usually encompasses just single-value annotation or recording of the d33 value in a lab book. Optionally, there is option of connecting the PM300 to a PC to dump measurement data on the latter –this is an option barely needed. May the user refer to the PM300 operation manual for details.

Indirect extraction of d33 is one of the alternate methods used to benchmark the PM300-extracted d33 values–e.g. by means of electrical characterization of the frequency response of the resonant and antiresonant frequencies of a piezoelectric device. However, this SOP suggests the use of specific test/reference structures with known/expected d33 values, and that can be used in combination with the PM300 vendor's calibration.

4. EQUIPMENT AND WORKPLACES

Equipment layout (room plan)

Infrastructure

Equipment list (name, no. of person responsible) - if desired

Equipment-specific hazards and safety measures (incl. pictograms)

Workplace-specific hazards and safety measures (incl. pictograms)

Media supply and disposal

Chemical hazards incl. gases

To be completed

5. RISK ASSESSMENT

A risk assessment according to MSR is to be carried out for self-produced machines or machine parts. (if required) *SPECIFIC RISKS OF RUNNING THIS PROCEDURE / TOOL

To be completed

6. EMERGENCIES AND ACCIDENTS

Reporting obligations - Reporting channels

Fire protection - Special laboratory area

First aid - Special laboratory area (chemical, thermal, mechanical)

Escape and rescue plan (general plan)

Emergency stop system (if available)

Emergency shut down system (if available)

7. RECORDS AND DOCUMENTATION SAFEKEEPING

The associated PM300 RA XXX-C&S-EqRA-274 is stored on QT9 and must be read, electronically signed off by all users. QT9 also stores a log of all users who have completed this risk assessment. PM300 user manual and specification document are also available on QT9.

8. RESPONSIBLE PERSONS

It is the responsibility of the HoRU and relevant PM to ensure the user has read and signed off the risk assessment for PM300 PiezoMeter Risk Assessment XXX-C&S-EqRA-274 and completed OACF C.1.11 SOP for Lab Induction in advance of using the PM300 PiezoMeter.
It is the responsibility of all users to read the user manual (located in the C.1.11 lab and on QT9) prior to using the PM300 Piezometer. It is the responsibility of all users to ensure strict compliance to this procedure.
It is the responsibility of the laboratory manager to ensure the user has read and signed off the C.1.11 OACF Lab RA Document XXX-C1.11-GRA-66.
It is the responsibility of the C.1.11 OACF laboratory manager to update this document, if required.

Creator (in block capitals): First name + Surname

Lab Manager (in block capitals): First name + Surname
Mail Adress plus Telephone number

Direct Supervisor (in block capitals): First name + Surname
Mail Adress plus Telephone number

....	....
Date, Signature Lab Manager	Date, Signature Supervisor

9. VERSION

Version	Date	Comment	Prepared/amended by
V1.0	XX.YY.ZZ	Creation	XXXXXXXX

To be completed